13 $\sqrt{5}(\sqrt{8} + \sqrt{18})$ can be written in the form $a\sqrt{10}$ where a is an integer.

Find the value of a.

×	√8	√18
√5	√40	√90

1 mark
Multiply the surds

$\sqrt{40} + \sqrt{90} = \sqrt{(4 \times 10)} + \sqrt{(9 \times 10)}$

1 mark
Extract the square factors

$= 2\sqrt{10} + 3\sqrt{10} = 5\sqrt{10}$

1 mark
The coefficient is 5

$a =$ 5

(Total for Question 13 is 3 marks)